VikSetu

Civic Data Intelligence Platform: ingest, analyze, map, and AI-assisted exploration of government open data.

Big Data project | Multi-language documents (English, Hindi, Marathi) | Pilot: Palghar FY 2024-25

1. Introduction

Local governance produces budgets, annual reports, scheme lists, and facility registries, but these sources use different formats: digital PDFs, scanned Marathi documents, web pages, and portal exports. VikSetu automates organization, extraction, and analysis, then presents results on an interactive map with charts and a chat assistant that controls the interface through validated tools.

2. Problem Statement

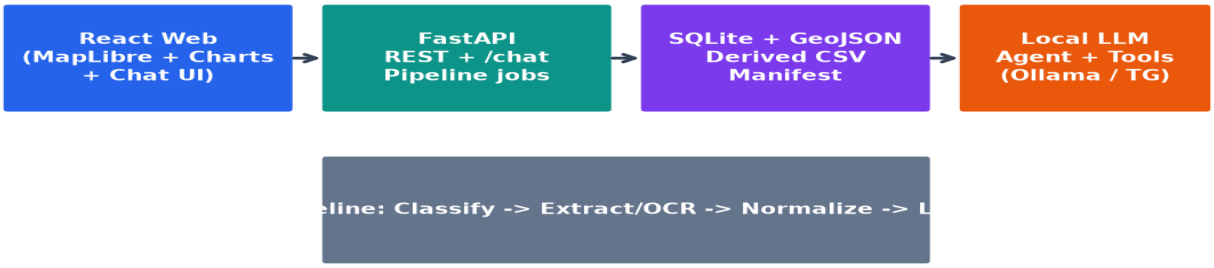
Citizens and administrators struggle to obtain a single view of allocation versus expenditure by sector and region, to locate facilities relative to tribal and rural population, and to trace figures back to official documents. Manual spreadsheet work does not scale and is error-prone for Devanagari tables.

3. Objectives

- Ingest heterogeneous civic files via a workspace with automatic classification.
- Extract fiscal and scheme tables into SQLite with full provenance metadata.
- Visualize Palghar on a MapLibre dashboard with linked KPIs and charts.
- Deploy a local-first ReAct agent; optional cloud LLM for OCR/table repair only.
- Provide audit-friendly traces for viva and project evaluation.

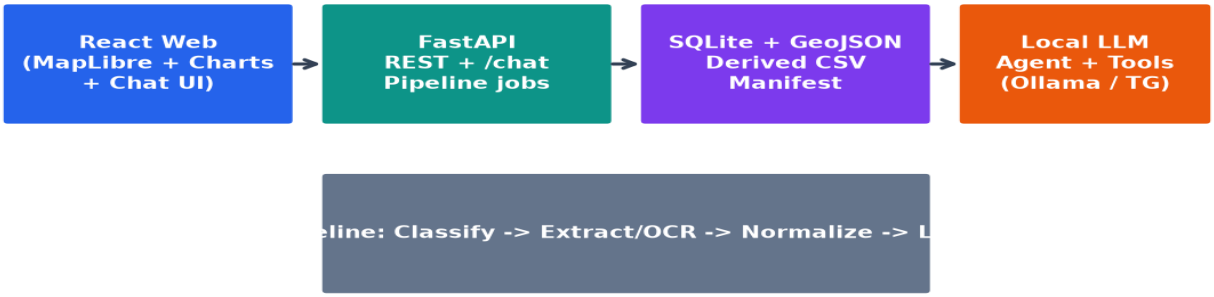
Figure 1: High-level architecture (see Page 2 for detail).

Figure 1: VikSetu three-tier system architecture



4. System Architecture

Figure 1: VikSetu three-tier system architecture

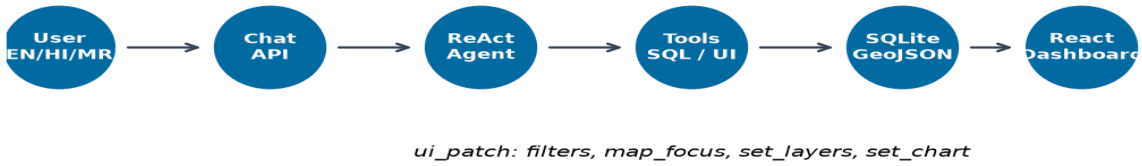


The frontend is a React single-page application. MapLibre renders taluka polygons and optional facility pins. ECharts or Recharts render sector bars and status donuts. A shared filter state synchronizes map, table, and charts. The FastAPI backend serves GeoJSON layers, spend aggregates, upload endpoints, and the chat route. Python hosts the pipeline and extends the training ReAct agent with VikSetu-specific tools.

5. AI and Language Strategy

One multilingual orchestrator LLM (local Ollama or text-generation-webui) handles chat in English, Hindi, or Marathi, document classification, and JSON tool calls. Tesseract OCR handles bulk scanned text (mar+eng+hin). Optional Gemini Flash acts as backup for difficult table pages. Numeric answers always originate from SQL on SQLite views, not from free-form generation. Lightweight RAG may explain narrative sections but does not compute totals.

Figure 5: Site-connected chatbot control flow



ui_patch: filters, map_focus, set_layers, set_chart

6. Why Not RAG-Only?

Retrieval-augmented generation splits tables across chunks and hallucinates amounts. VikSetu uses structured ETL plus tool calling, which aligns with Big Data learning outcomes: preprocessing, storage, analysis, visualization, and optional prediction on derived series.

7. Data Processing Pipeline

Figure 2: Multi-AI data processing pipeline



Drag-and-drop workspace triggers N0-N10 | v1: fixed chain with progress stepper

- N0 Ingest: copy uploads to inbox/; compute hash; append manifest.json.
- N1 Classify: doc type, language, region (Palghar keywords), needs_ocr flag.
- N2/N3 Extract: PyMuPDF text path or Tesseract OCR for scanned ZP budget PDF.
- N4/N5: Table recovery and normalize to spend_lines / works schema.
- N6-N8: Region bind, village/taluka geocode, load SQLite and geo/ exports.
- N9-N10: Precompute insights.json; expose via REST for dashboard and agent.

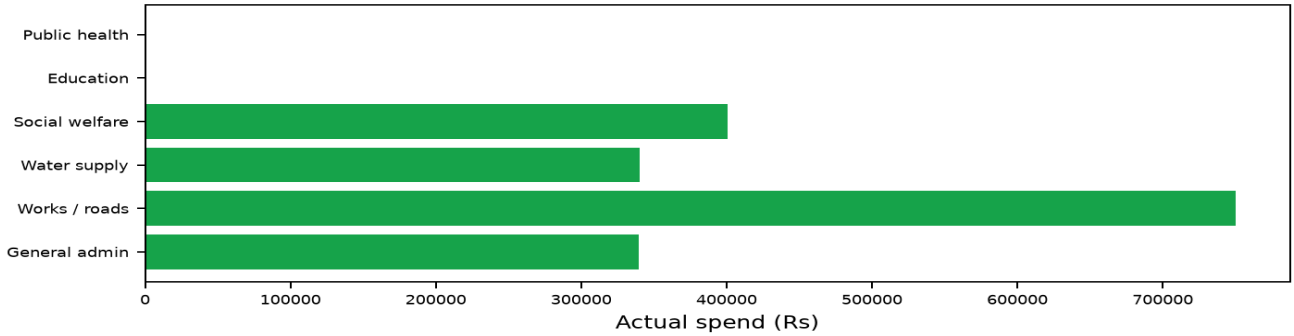
8. Data Sources (Pilot)

- Zilla Parishad Palghar budget 2024-25 (scanned, ~10 MB) and schemes web pages.
- Panchayat Samiti Palghar annual administration report 2024-25 (text PDF, validated).
- Maharashtra DES indicators; MGNREGA MIS; eGramSwaraj; MRSAC Health GIS; OSM POIs.

Sample Extract: PS Panchayat Samiti Spend FY 2024-25 (Rs)

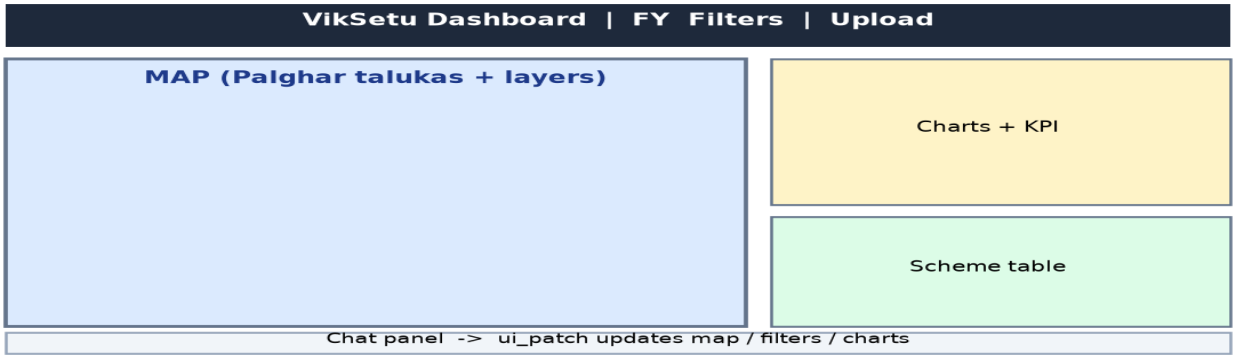
Sector	Allocation	Actual spend
General admin	340,000	339,638
Works / roads	750,000	749,986
Water supply	340,000	340,000
Social welfare	440,000	400,680
Education	0	0
Public health	0	0
Total	**2,200,000**	**1,939,204**

Figure 4: Sector spend from PS report FY 2024-25 (pilot data)



9. Geospatial and Dashboard Design

Figure 3: Map-first web dashboard layout



- Layers: 8 talukas (choropleth spend or utilization), health/school pins, works by status color.
- Filters: financial year, taluka, sector, status (completed / in progress / planned / no spend).
- Linked views: selecting a chart category highlights map features; table sorts by amount.
- Chat ui_patch applies the same filter object as manual UI controls for consistent demos.

10. Agent Tools (Autonomous Behaviors)

- query_spend, list_works: read-only SQL with document citations.
- set_filters, map_focus, set_layers, set_chart: dashboard control without DOM hacking.
- show_source: open provenance panel for examiners.
- Pipeline on upload: classify region, route OCR, reload derived tables.

11. Five-Day Build Plan

- Day 1: SQLite + API + taluka GeoJSON.
- Day 2: React map and charts with filter state.
- Day 3: Agent /chat and ui_patch; scripted demo questions.
- Day 4: Upload stepper and pipeline demo; facility layer if time permits.
- Day 5: Documentation, PDF, video, README.

12. Limitations and Ethics

- Pilot uses authentic PS extract; full ZP OCR may be partial within deadline.
- Geocoding may remain taluka-level until village gazetteer matching is complete.
- Public data only; helper tool, not statutory audit or RTI replacement.

13. Success Metrics

- Real FY 2024-25 totals displayed from database, not hard-coded in UI.
- Map renders Palghar talukas; chat changes filters and map focus live.
- Upload runs classify/extract with logged trace.

14. Deliverables

- viksetu-api, viksetu-web, pipeline scripts, agent tools, data/ folder.
- This document, implementation notes, demonstration recording.

15. Conclusion

VikSetu bridges fragmented government records and actionable spatial analytics. The Palghar pilot demonstrates real expenditure patterns (for example works and water spending with zero education/health spend in the PS consolidated extract). The architecture scales to additional districts by repeating ingest and region binding without redesigning the agent or dashboard core.

16. References

- zppalghar.gov.in; Panchayat Samiti Palghar annual report PDF.
- mahades.maharashtra.gov.in; nregastrep.nic.in; egramswaraj.gov.in.
- MRSAC Health GIS; Open Budgets India; Datameet / OSM boundaries.
- Training modules: agent.py, tools.py, llm.py (ReAct pattern).

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